REVIEW

Particle & Particle Systems Characterization
www.particle-journal.com

Metal Hydride Storage Systems: Approaches to Improve Their Performances

*Wei Liu, Joseph Almar Tupe, and Kondo-Francois Aguey-Zinsou**

Metal hydrides provide a safe and efficient way to store hydrogen. However, current metal hydride storage systems, i.e., hydrides incorporated within a storage tank, are far from efficient. Depending on the design, (dis)charging rates may be very long. However, this can be significantly improved by implementing strategies tackling the issue of heat management at the level of: i) the metal hydride bed, and ii) the overall storage system design. This review summarises recent progress in tackling heat management of hydride systems. In this respect, modeling has emerged as a powerful tool. In particular, simulation results show that the compaction of hydride powders with binders and the use of metal foams are both effective in lifting the poor thermal conductivity of hydride beds. For tank designs, cylindrical shapes remain the preferred choice because of the flexibility and ease of supplementing heat management with fins and tubular heat exchangers. The addition of phase change materials to the hydride tank can lead to further heat storage, but any add-on to simple hydride tanks can only lead to cumbersome systems. It is still a fine art to tune the thermal conductivity of hydride beds while selecting a suitable metal hydride alloy composition.

1. Introduction

Hydrogen is a clean energy carrier with a high energy density of 142 MJ kg^{-1} and has the potential to replace fossil fuels to decarbonize society.[1] However, the transition to hydrogen as the main energy vector is challenging because there are still barriers to large-scale application in hydrogen production, storage, and transport. Hydrogen storage is considered a bottleneck because hydrogen has low density under ambient conditions, and this leads to low volumetric energy densities. It is more than ever critical to develop storage technologies that can store hydrogen in a safe and compact form.

There are currently three main approaches to storing hydrogen, e.g. high-pressure gas storage, cryogenic temperature liquid storage, and metal hydrides solid storage.[2] Compressing hydrogen at 15–70 MPa is the most mature technology. However, it is of low volumetric density (10–40 kg m^{-3}).[3] Liquefying hydrogen increases the volumetric density to ≈70 kg m^{-3}, but hydrogen liquefication is energy intensive with an energy consumption of 12.0–15.5 kWh per Kg of H_2.[4] In this case, hydrogen boiling-off is also a problem leading to safety issues. The alternative approach is to use metal hydrides storing hydrogen under mild conditions, e.g. near or at room temperature. Over the years, many metal and complex hydrides have been developed (**Table 1**); and depending on application requirements, hydrides with specific properties can be selected, e.g. LaNi-based alloys of 1.2 wt% H_2 capacity at room temperature, MgH_2 with a 7.6 wt% H_2 capacity with hydrogen uptake/release possible > 300 °C, and $NaAlH_4$ of 5.6 wt% H_2 reversible capacity with an operating temperature ≈200 °C.[5]

In addition to hydrogen storage, metal hydrides also have been used for hydrogen separation and purification.[6] In this process, gas mixtures can be filtered by metal hydrides forming beds releasing only hydrogen. The potential of metal hydrides also expands to the compression of hydrogen owing to the dependence of the equilibrium plateau pressure on temperature,[7] and applications including heat storage and heat pump since the hydrogen absorption reaction in metal hydrides is exothermic and the desorption process is endothermic.[1] For all these applications, the metal hydride bed must be enclosed with a tank that can be pressurized to control the hydrogen absorption and desorption process, while enabling an effective exchange of heat between the hydride bed and the tank surroundings.

Hereafter, a metal hydride bed means a hydrogen storage material in the form of a powder or pellets plus an additive to increase its thermal conductivity, while the storage tank refers to the container incorporating the hydride bed plus any features to increase thermal conductivity, e.g. fins or heat exchangers. A hydride hydrogen storage system is thus one or multiple storage tanks. The performance of a hydride hydrogen storage system is mainly defined by the amount of hydrogen it can store

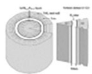
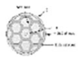
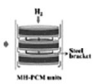
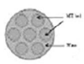
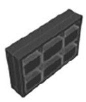
W. Liu, J. A. Tupe, K.-F. Aguey-Zinsou
MERLin
School of Chemistry
University of Sydney
Sydney, NSW 2006, Australia
E-mail: f.aguey@sydney.edu.au

The ORCID identification number(s) for the author(s) of this article can be found under https://doi.org/10.1002/ppsc.202400163

DOI: 10.1002/ppsc.202400163

Particle & Particle Systems Characterization
www.particle-journal.com

Table 1. Summary of reported metal hydride storage systems performance.

Storage materials	Tank design	System description	Charging rate [kg H_2 per min]	Refs.
10 kg $LaNi_5$		Cooling tubes External water jacket	0.004	[26]
110 g $LaNi_{4.25}Al_{0.75}$		Double-layered reactor Cooling tubes N_2 as heat transfer fluid	<0.001	[27]
50 kg $La_{0.9}Ce_{0.1}Ni_5$		Honeycomb hexagonal fins	0.005	[28]
5 kg $LaNi_5$		Internal conical fins Embedded cooling tubes	0.008	[29]
10 kg MgH_2		With phase change material	0.029	[17d]
210 kg Ti—Mn		Multi-tubular storage	0.078	[30]
$LaNi_5$		Dual coil cooling tube	0.001	[31]
$LaNi_5$		Compartmentalised reactor	0.001	[25]
$LaNi_5$		Concentric finned tube	0.003	[32]
1 kg $LaNi_5$		Water outer jacket, Embedded spiral cooling tube	0.001	[33]
100 kg $NaAlH_4$		Cooling tubes interconnected by aluminum fins	0.272	[34]

Particle & Particle Systems Characterization
www.particle-journal.com

and the rate at which this amount can be absorbed and released. In most cases, the hydrogen storage capacity is dependent on the storage material being used. However, the rate of charging and discharging is influenced by many different factors, including the properties of the storage material, the tank geometry, and the operating parameters. As mentioned above, there is heat release and uptake along with the interaction of hydrogen gas (H_2) with the metal (M) forming a hydride (MH_x) following Equation 1.

$$M\,(s) + \frac{x}{2}H_2\ \ (g) \rightleftharpoons MH_x\,(s) + \Delta H \tag{1}$$

Therefore, the optimal operation of hydride hydrogen storage systems is based on efficient heat management during the hydrogen absorption and desorption reactions. Since both processes are based on the same physical-chemical principles, all the passive elements implemented in hydride beds and tanks equally contribute to the heat-transfer rate of the overall system and, as such, the hydrogen uptake/release performance of the metal hydride system. The absorption process is an exothermic chemical reaction that releases a significant amount of heat (ΔH). If not dissipated in a timely manner, this heat eventually displaces the chemical equilibrium back toward the reaction promoting hydrogen release, and this can severely impact the performance of the metal hydride tank. Equivalent chemical principles affect the desorption process, but now heat should be provided to the tank to favor the release of H_2 as a gas.[8] As such the hydrogen absorption and desorption rates slow down upon ineffective heat management.[9] Recent efforts in developing efficient thermal management systems have been geared toward improving the hydrogen absorption rate in hydride hydrogen storage systems.[10] In this respect, modeling and simulation have emerged as powerful tools. Mathematical models featuring heat conduction laws, for example, Computational Fluid Dynamics (CFD), have become indispensable tools in the design of hydride tanks because of the ease of assessment of the heat transfer performance.[11]

In the 1980s, basic mathematical models,[9,12] which are the foundations of later studies, were developed to simulate heat and mass transfer within metal hydride beds.[13] Over the years, great efforts have been devoted to improving these models and bringing them closer to realistic physical conditions. In this respect, many variable parameters can now be simulated to comprehend the performance of hydrogen storage systems, including the hydride bed's physical/chemical properties, the tank design, and heat exchanger configurations.[14] However, although various heat exchangers have been proposed, there is still no efficient approach to achieve high hydrogen (dis)charge rates without sacrificing the overall hydrogen storage capacity of the tank. Several reviews have already summarised approaches to thermal management for hydrogen storage systems,[14,15] but there is still a lack of systematic reviews integrating the basic mathematical equations, the storage materials, hydride beds, tank shapes, and tank material with the approaches to improve the overall performance of hydrogen storage systems. This review also provides a good tutorial to guide correct modeling techniques for hydride tanks. The first section of this review is an introduction to the mathematical models required. Then, in combination with examples of current simulation capabilities, methods to improve the performance of hydride hydrogen storage systems are discussed.

2. Modelling in Metal Hydride Hydrogen Storage System

Modeling has increasingly attracted interest in metal hydride system design. In this section, the principles of heat conduction and associated mathematical models that can be used to facilitate this are introduced.

2.1. Principles of Heat Conduction: Fourier's Law

Heat conduction is the process by which two systems exchange energy in the form of heat transferred through a material due to an existing temperature gradient. The rate at which heat is exchanged (Q) across one direction in any material system can be calculated by using Fourier's law of heat conduction (Equation 2).[16]

$$Q_{1\to 2} = -k\frac{A\Delta T_{1\to 2}}{d} \tag{2}$$

The heat transfer rate ($Q_{1\to 2}$) between two points labelled 1 and 2 is directly proportional to the cross-sectional surface area (A) perpendicular to the direction of heat flow, the temperature gradient between the initial and final points ($\Delta T = T_2 - T_1$), and the thermal conductivity (k) of the material that acts as a medium. It is also inversely proportional to the distance (d) between points 1 and 2 (**Figure 1**).

Most of the designs proposed in the literature aim to maximize the value of Q without compromising the total weight and volume of the hydrogen tank and manufacturing costs. Some examples of these designs introduce fins to simultaneously increase the surface area of the tank in contact with the bed, add heat exchangers to maximize the temperature gradient, or use chemical additives in the alloy bed to enhance the thermal conductivity of the hydride.

2.2. Mathematical Models

As mentioned above, CFD is the preferred computational method to test the designs of these systems. Most of the work developed in the field over the last 25 years has used slight variations of the mathematical model presented below that is built on the following main assumptions:[17]

- Hydrogen is the only fluid, and it behaves like an ideal gas;
- The hydride is considered as an isotropic bed with homogeneous porosity and no change in volume;
- The main physical properties remain constant during the reaction;
- Local thermal equilibrium between the hydride bed and the hydrogen gas is assumed.

2.2.1. Mass Balance: Thermodynamic and Kinetic Models

The mass conservation equation is the only duplicate conservation equation in the model. It simultaneously captures the

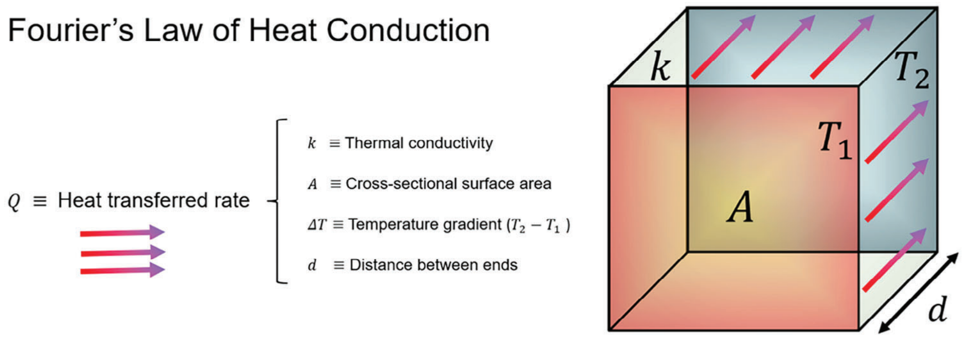


Figure 1. Graphical representation of the Fourier's law of heat conduction.

depletion (release) of hydrogen gas in the tank and the formation of the hydride (alloy) during the absorption (desorption) reaction. The mass balance for the hydrogen gas is represented in Equation 3.

$$\varepsilon \frac{\partial \rho_g}{\partial t} + \nabla \rho_g \vec{v} = -\phi_i,\ i \equiv abs,\ des \tag{3}$$

where ϵ is the porosity of the bed. The symbol ϕ_i is the source term that results from the kinetics of the absorption or desorption reactions. The variations in the gas density ρ_g in the chamber are subjected to changes in pressure ($p = p_{H_2}$) and temperature (T) following the ideal gas equation of state (Equation 4):

$$\rho_g = \frac{pM_{H_2}}{R_g T} \tag{4}$$

Similarly, for the hydride (solid), the mass conservation is as follows (Equation 5):

$$(1-\varepsilon)\frac{\partial \rho_s}{\partial t} = \phi_i \tag{5}$$

The source term ϕ_i for the absorption and desorption models (Equations 6 and 7) are derived from kinetic studies in metal hydrides,[12a] and are generally valid for different alloy compositions:

$$\phi_{abs} = c_{abs}\exp\left(\frac{-E_{abs}}{R_g T}\right)\ln\left(\frac{p}{P_{eq,\,abs}}\right)\left(\rho_{s,sat} - \rho_s\right) \tag{6}$$

$$\phi_{des} = c_{des}\exp\left(\frac{-E_{des}}{R_g T}\right)\left(\frac{p - P_{eq,des}}{P_{eq,des}}\right) \tag{7}$$

E_i and c_i are, respectively, the activation energy and the kinetic rate constant of each absorption and desorption reaction, $\rho_{s,sat}$ is the hydride concentration at the end of the reaction (fully loaded system), and $P_{eq,i}$ is the equilibrium pressure of the chemical reaction. $P_{eq,i}$ is generally implemented in the model by fitting the pressure-composition-temperature (PCT) curve to a convenient function valid within the range of operating conditions.[18] In most alloys, $P_{eq,abs}$ differs from $P_{eq,des}$ due to their intrinsic hysteresis.[19] Different approximations can be used for simulating the equilibrium plateau pressure of the hydride,[20] the most commonly used and employed is the function proposed by A. Jemni and S. Ben Nasrallah (Equation 8).[17a,b]

$$P_{eq,i} = \sum_{j=1}^{n=9} a_j (H/M)^j \exp\left(\frac{\Delta H_i}{R_g}\left(\frac{1}{T} - \frac{1}{T_0}\right)\right) \tag{8}$$

In (8), $P_{eq,i}$ is a function of the level of hydrogen concentration expressed as the hydrogen-to-metal ratio (H/M), the temperature of the bed T, and the enthalpy of formation ΔH_i.

2.2.2. Momentum Conservation and Energy Equation

In fluid mechanics, the momentum balance equation, derived from Newton's second law of motion, states that the rate of change of momentum of a fluid is equal to the sum of the forces acting on that element. In this case, the momentum balance equation is as follows:

$$\frac{\partial \rho_g \vec{v}}{\partial t} + \nabla \rho_g \vec{v}\vec{v} = -\nabla p + \nabla \bar{\tau} + \phi_D \tag{9}$$

The momentum source term ϕ_D is the expression of Darcy's law[21] to reproduce the diffusion of gas through a homogeneous porous media, i.e., the metal hydride bed.

$$\phi_D = -\frac{K}{\mu}\nabla p \tag{10}$$

K is the permeability, μ the viscosity of hydrogen gas, and ∇p the drop in pressure. The Kozeny-Carman equation permeability of the bed related to the porosity (ϵ) and particle size (d_p) is given by:[22]

$$K = \frac{d_p^2}{150}\frac{\varepsilon^3}{(1-\varepsilon)^2} \tag{11}$$

The general expression of the energy equation for gas and solid are respectively:

$$\varepsilon \rho_g C_{p,\,g}\frac{\partial T^g}{\partial t} = \varepsilon k_g \nabla^2 T^g - \rho_g C_{p,\,g}\left(\vec{v}\nabla T^g\right) + \phi_{Q^g} \tag{12}$$

$$(1-\varepsilon)\,\rho_s C_{p,\,s}\frac{\partial T^s}{\partial t} = (1-\varepsilon)\,k_s \nabla^2 T^s + \phi_{Q^s} \tag{13}$$

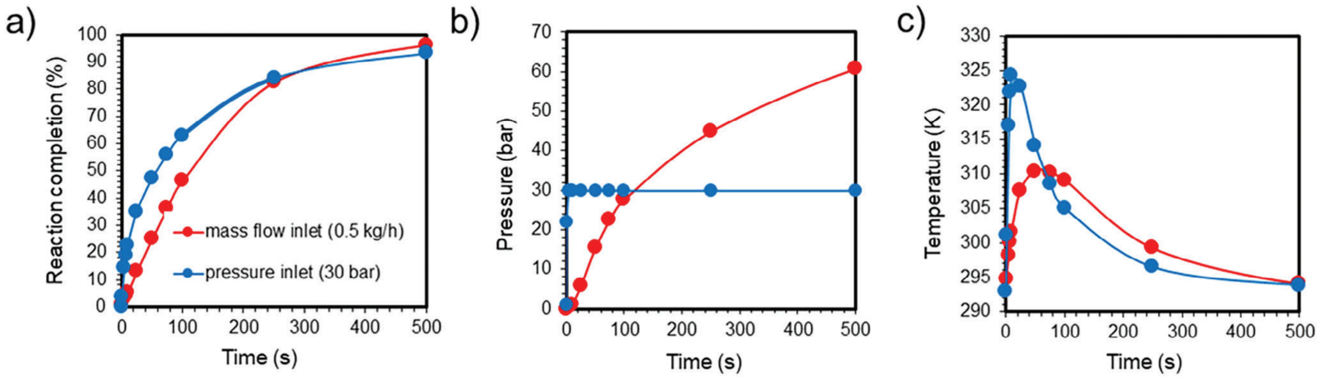


Figure 2. Comparison of simulation results of a hydride tank with the hydrogen pressure or mass flow at the tank inlet set as a fixed parameter. a) Hydride formation rate. b) Pressure evolution in the hydride tank. c) Temperature evolution in the hydride tank. The initial tank temperature was set at 293 K and the interior pressure of the tank was set at 0.01 MPa.

As a side note, this energy equation is the mathematical formulation of the law of conservation of energy. It states that the rate at which energy enters the volume of a moving fluid is equal to the rate at which work is done on the surroundings by the fluid within the volume and the rate at which energy increases within the moving fluid.[23]

The assumption of the local thermal equilibrium between the gas and the hydride bed (i.e., $T^g = T^s$) allows to condensing of the individual specific heat capacities and thermal conductivities of the gas and the solid into a single effective specific heat capacity $(\rho C_p)_{eff}$ and a single effective thermal conductivity (k^{eff}):

$$\left(\rho C_p\right)_{eff} = \varepsilon \rho_g C_{p,\,g} + (1-\varepsilon)\,\rho_s C_{p,\,s} \tag{14}$$

$$k_{eff} = \varepsilon k_g + (1-\varepsilon)\,k_s \tag{15}$$

Now, Equations 12 and 13 can be merged into a single energy equation.

$$\left(\rho C_p\right)_{eff} \frac{\partial T}{\partial t} = k_{eff} \nabla^2 T - \rho_g C_{p,\,g} \left(\vec{v}\nabla T\right) + \phi_Q \tag{16}$$

With the heat source term, that depends on the reaction rate (ϕ_i) and is valid for both absorption and desorption reactions:

$$\phi_Q = \phi_i \left(\frac{\Delta H_i}{M_{H_2}} + T\left(C_{p,\,g} - C_{p,\,s}\right) \right) \tag{17}$$

2.2.3. Initial and Boundary Conditions

All modeling requires initial and boundary conditions. Initial conditions are used to define the initialization of various parameters in the studied system, while the boundary conditions define how a system interacts with the environment.[24] In the simulated case below (**Figure 2**), looking at cylindrical tanks with a single H_2 inlet at the top, and a total volume of 0.5 L. The boundary conditions, for example, assume a top 1/5 of the volume of the tank as vacuum or occupied only by hydrogen gas, while the bottom 4/5 is made of a porous bed of $LaNi_5$ ($\epsilon = 0.5$). Convective air (293 K) is used as the external tank conditions separated from the inside by virtual stainless steel walls exchanging heat with the interior using the following expression.

$$q = h_f \left(T_{ext} - T_{int}\right) \tag{18}$$

Under these conditions, two methods can be used to control the entry of hydrogen at the input: i) as a set pressure (e.g., 3 MPa), or as a set mass flow (e.g., 0.5 kg h^{-1}). A computational study reveals similar reaction completion times under both conditions (Figure 2). When a set pressure is used, the tank is quickly pressurized, and the absorption rate becomes remarkably high in the initial states of the simulation, leading to a sharp increase in temperature. When a constant mass flow is used, the tank fills up at a constant rate, and the temperature takes more time to increase without reaching values as high as those obtained using a set hydrogen pressure at the tank's inlet. Because the pressure is not fixed, it can reach high values of ≈6 MPa in 500 s because of the lack of boundary limit on tank pressure when a set hydrogen mass flow is used.

Accordingly, fixing the pressure at the input is a good method for studying the effectiveness of innovative designs in temperature dissipation. Fixing the mass flow rate at the inlet can help realistically model the operation of the tank under real conditions.

3. Current Performance of Metal Hydride Storage Systems

For hydrogen storage systems, the US DOE has set 2020 targets of 0.5 kg min^{-1} H_2 charging/discharging rates for material handling equipment. Current charging rates of some designed metal hydride hydrogen storage systems are listed in Table 1, and most reported systems are far from achieving such rates. Some demonstration and application projects are also listed in **Table 2**.

It is simple and effective to improve the hydrogen storage performance by dividing the storage tank into subsections or compartments. For example, a 50% reduction in charging time for a rectangular $LaNi_5$ tank was observed when the tank was divided into six equal compartments.[25] However, this increases the overall weight and volume of the hydrogen storage system.

Table 2. Demonstration and application of hydride-based hydrogen storage systems.

Project or manufacturer	Alloy	Hydrogen amount [kg]	Pressure [bar]	Heat management	Refs.
HyCare	Metal hydride	50	50	–	[38]
GKN Hydrogen	Metal hydride	10–25, 30–120, 250	40	–	[39]
Ovonic Hydrogen Systems	Ovonic alloys	Up to 10 kg	103–345	–	[40]
BOR4STORE	$LiBH_4$ and MgH_2	0.04–0.05	3–100	Multi-module with cooling jacket	[41]
H2Fuel Project	1250 kg $Lm_{1.06}Ni_{4.96}Al_{0.04}$	15	<35	Al foam, U-shaped water tube	[42]
THEUS	50 kg $MmNi_5$	0.57			[43]

Similarly, a study on a storage system with a honeycomb metallic structure compartmentalizing $NaAlH_4$ found that the hydrogen charging time was controlled by the cell wall length. In this case, the amount of stored hydrogen increased by 25.4% for a smaller cell wall length of 0.952 cm.[35] Another strategy that has been gaining a lot of attention is to assemble smaller metal hydride reactor units into larger industry-scale systems. A system bundling four tubes of 25 kg of $MmNi_{4.6}Al_{0.4}$ each, achieved a hydrogen charge time of 600 s,[36] while a storage system with 210 kg Ti—Mn alloy distributed across 14 cylindrical tubes enclosed in a large tank containing the cooling fluid completed the absorption process in 900 s and the desorption process in 2000 s.[30] A similar concept was investigated where 16 metal hydride tubes were stacked in parallel to achieve an 8 kg $NaAlH_4$ system. This design was predicted to be refuelled at 80% full capacity in 209 s.[37]

Other various approaches, e.g. fins,[29,30,32] cooling tubes,[26,27,29,31–33] phase change materials,[17d] have also been investigated in an attempt to improve heat management. However, the hydrogen charging rate of these systems was still lower than the DOE technical target for material handling equipment. The best performance of 0.272 kg H_2 min^{-1} was reported with $NaAlH_4$ in a tank with cooling tubes interconnected by aluminum fins. The latter are expected to facilitate the heat transfer within the hydride bed.[34] To reach higher hydrogen refilling rates, novel approaches leading to better thermal conductivity should be developed.

4. Approaches to Improve the Performance of Hydride Storage Systems

As discussed above, to improve the performance of hydrogen storage systems, not only the hydride beds but also the tank design needs to be improved. In terms of a hydride bed, the storage material properties, bed form, additive or matrix in the bed are all important parameters. For tank design, the nature of the tank material, its shape/dimensions, fins, and the nature of the type of heat exchangers are additional levers to improve heat management.

4.1. Improving the Thermal Conductivity of Hydride Beds

4.1.1. Low Thermal Conductivity of Hydride Beds in Powder Form

As mentioned above, many hydrogen storage materials with different properties have been developed. Storage properties and thermal conductivity values of common storage materials are summarised in **Table 3**. Suitable storage materials can be selected according to the application requirements.

Alloys forming hydride have a relatively poor thermal conductivity, and upon hydrogen cycling, the hydride particles are subjected to strain changes, leading to extended decrepitation of the hydride powder and poor interparticle contact.[44] For example, the thermal conductivity of inactivated $LaNi_5$ ingots and pulverized $LaNi_5$ powder upon the first hydrogen activation was measured to be 1.63 and 0.84 W m^{-1} K^{-1}, respectively.[45] Therefore, these alloys usually show a low effective thermal conductivity between 0.1 to 1.5 W m^{-1} K^{-1} depending on the operating temperature and hydrogen pressure (Table 3).[1,46] This poor thermal conductivity of metal hydride powders significantly dictates the heat transfer resistance and thereby impacts the hydrogen charge–discharge time.[47]

Several investigations on storage tanks with alloy powders of different particle sizes reached the same conclusion, i.e., the alloy particle size has no significant effect on the fill time. For example, study of a tank with $LaNi_5$, $LaNi_{4.75}Al_{0.25}$, or $MmNi_{4.6}Al_{0.4}$, showed that particles of 0.01, 0.1, 0.5, and 1.0 mm in size led to the same hydrogen absorption times.[49] Similarly, $LaNi_5$ of 20 and 60 µm in the storage system showed the same internal temperature and fill rates of 500 s for 120 g of alloy.[50] However, an investigation of a system containing the AB_2 alloy $Ti_{0.98}Zr_{0.02}V_{0.43}Fe_{0.09}Cr_{0.05}Mn_{1.5}$ claimed that the smaller particles resulting from the pulverization of the alloy increased the reaction rate constant from 0.02 to 0.56 s^{-1}, while the thermal diffusion decreased from 1.1834×10^{-6} to 1.1285×10^{-6} m^2 s^{-1}. The thermal diffusivity of the bed remains the main factor affecting

Table 3. Properties of hydrogen storage materials relevant to hydride tanks.[1,46a,48]

Material	Capacity [wt%]	Operation pressure [MPa]	Operation temperature [°C]	Thermal conductivity [W m^{-1} K^{-1}]
$LaNi_5$	1.3	0.001–1	20–60	0.1–1.5
$LaNi_{4.7}Al_{0.3}H_x$	1.2	0.001–6	−20–140	0.1–1.1
$TiFeH_x$	1.9	10	25	1.49
$Ti_{1.1}CrMnH_x$	1.8	0.29–25.3	20–27	0.3–0.7
$TiMn_{1.5}H_x$	1.8	0.05–5	22–60	0.25–1.2
$NaAlH_4$	5.6	0.1–5	30-200	0.25–1.2
MgH_2	7.6	0.02–1.5	250–380	2–8
$LiBH_4/MgH_2$	10.0	0.3–10	max. 650	0.9–1.6

the overall system performance when the hydrogen absorption rate of the metal hydride tank is lower than the intrinsic hydrogen sorption rate of the material.[51]

4.1.2. Hydride Bed Modifications

In order to improve the thermal conductivity of metal hydride beds, three main methods have been investigated. These include compacting the metal alloy into pellets with binding additives, the use of metallic matrices, and/or metal coating/encapsulations.

Compacted Pellets with Additives: One of the effective methods to improve the alloy bed's thermal conductivity is to increase the contact area between the particles of the hydrogen storage material by compression to form a dense pellet. However, too dense pellets could lead to a deterioration of hydrogen permeability due to the reduced pellet porosity. Therefore, in most cases, an additive, e.g. aluminum (Al) powder,[53] or expanded natural graphite (ENG)[54] is added during compression to increase the thermal conductivity while binding the hydride particles.

Aluminum is cheap and has a high thermal conductivity of 247 W m^{-1} K^{-1}.[1] A compact can be formed by cold pressing the hydride forming alloy powder and the Al powder. This is usually followed by a sintering step at 150–250 °C and 20–30 MPa H_2 pressure with the aim of forming a porous structure that is stable upon hydrogen cycling.[53c] In practice, the optimized volume fraction of Al should be between 0.15 and 0.5 of the bed to reach a thermal conductivity of 8 to 23 W m^{-1} K^{-1}, as determined for $LaNi_5$-Al pellets.[53a] Such Al-based pellets showed good mechanical stability over 30000 hydrogen absorption-desorption cycles. However, the addition of Al reduces the hydrogen storage capacity by a factor of 0.3.

ENG is also a conventional binder used because of its low density and high thermal conductivity of 130 W m^{-1} K^{-1} (**Table 4**). Pellets formed by compressing the Hydralloy C52 ($Ti_{0.955}Zr_{0.045}Mn_{1.52}V_{0.43}Fe_{0.12}Al_{0.03}$) and 2.5, 5.0, and 12.5 wt% ENG showed good mechanical stability in comparison with pellets made of pure alloy. The former preserved their mechanical integrity after 85 hydrogen uptake/release cycles. More importantly, high radial thermal conductivity between 2.4–62.8 W m^{-1} K^{-1} was achieved depending on ENG amount and compression pressure, whereas the thermal conduction in the axial direction showed no change with the compression pressure and remained at ≈2 to 8 W m^{-1} K^{-1} (**Figure 3**). With ENG, the effect of compaction on the porosity was also reported. With an increase in compaction pressure, the porosity of the pellet decreased, e.g. from 34.9 vol% under 75 MPa to 15.7 vol% under 600 MPa for pellets with 12.5 wt% ENG. With an intermediate compression pressure of 75 MPa with 5 wt% ENG, an average thermal conductivity of 14.4 W m^{-1} K^{-1} and porosity of 31.1 vol% were obtained.[54b] Compared to the loose hydride powder, the pellets compacted at 75 MPa with 12.5 wt% ENG exhibited a 150% increase in volumetric storage capacity. It is noteworthy, that with ENG as an additive, no gases C-H compounds were reported to form upon hydrogen cycling, even during the high-temperature activation procedure at 300 °C.[54b]

Metal Hydride Beds Incorporating Metal Foams: Another commonly used method is to pack the hydrogen storage alloy into metal foams.[47,55] Metal foams have a high surface area, a porous structure, and high thermal conductivity. As such, these are good candidates to enhance the thermal conductivity of hydride beds. Al foams are the most popular, and studies showed that $LaNi_5$ beds with Al foams have the fastest hydrogen uptake rate in comparison to beds with copper (Cu) or zinc (Zn) foams (**Figure 4b**).[55a] The foam pore size and foam amount are also important factors to control. For example, a $LaNi_5$ bed with an Al foam with a pore size of 0.125 mm showed faster hydrogen absorption rates than an Al foam with a pore size of 0.2 and 0.35 mm (Figure 4c).[55a] Another study reported the same trend with $LaNi_5$ beds with foams of 2.3, 4.6, and 9.2 mm pore size, i.e., corresponding to porosities of 40 PPI (pores per inch), 20 PPI and 10 PPI, respectively.[56] In this case, the decrease in pore size leads to a higher heat transfer surface area per volume unit.[56] Therefore, pore sizes larger than 4.6 mm (porosity less than 20 PPI) are not recommended due to poor heat transfer.[56,57] It should be noted that smaller pore sizes increase the difficulty of packing hydrides within metallic foams.

Investigations on the effect of varying amounts of Al foams on the charging time showed that, obviously, the higher the number of Al foam blocks the better the rate of hydrogen uptake. For example, under natural convection charging times of $LaNi_5$ beds improved by 14 and 20% by increasing the amount of Al foam from 5 to 10 wt%, respectively.[55a] In a similar set-up but employing force convection, an enhancement of 32.0 to 41% was observed by increasing the amount of Al foam from 5 to 10 wt%.[55a] Overall, a 60 to 65% improvement in the charging time can be achieved when using an Al metal foam compared to a loose-packed bed powder (Figure 4d).[55a,58] For a certain amount of metal hydride, a mere 0.01 of metal foam volume fraction is reported to be sufficient to reduce the hydrogen charging time by a factor of six in comparison with that without metal foam,[59] while the optimal metal foam volume fraction was proposed to be of 0.08. Considering a $LaNi_5$ density of 8310 kg m^{-3}, a $LaNi_5$ porosity of 0.55, and a density of the Al foam of 2700 kg m^{-3}, the optimal Al foam amount in mass percentage is ≈5.9%.[55c] However, using metal foam has several disadvantages: i) it is hard to densely pack the storage alloy inside the pores of the foam, ii) this results in a loose contact between the hydrogen storage material particles and the foam's pore walls as wells as a lowered hydrogen storage capacities.[15b]

Table 4. Physical properties of related gases and materials for storage systems.[1,52]

Gases and materials	Thermal conductivity [W m^{-1} K^{-1}]	Tensile strength [MPa]	Ductility	Density [g cm^{-3}]
Hydrogen	0.17	–	–	–
Al	247	88	0.65	2.7
Al foam	11	–	–	0.4–1.2
Cu	398	220	0.62	9.0
Brass	109	300–330	0.05–0.55	8.4–8.7
Graphite	130	–	0.002	2.1
Tin	73	–	0.69	7.3
Ni	90	–	0.50	8.9
316 stainless steel	16	574	0.30–0.51	7.9

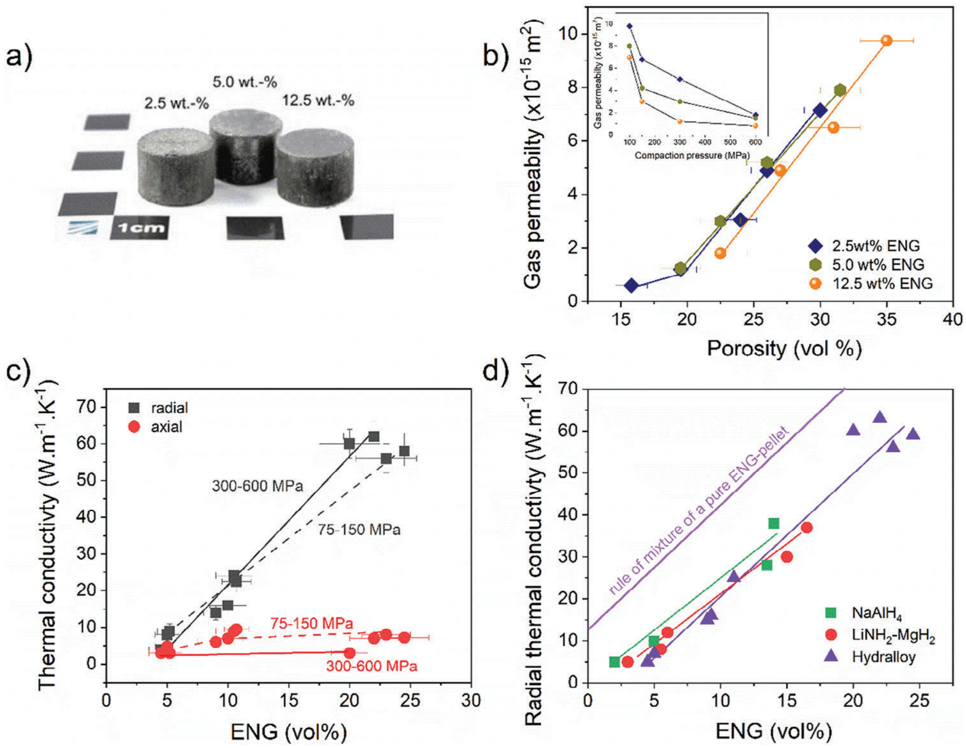


Figure 3. a) Hydralloy-ENG pellets with 2.5, 5.0 and 12.5 wt% ENG; b) Gas permeability versus compression pressure and porosity of the Hydralloy-ENG pellets; c) Thermal conductivities of the alloy-ENG pellets in the radial and axial direction with the alloy of Hydralloy; and d) Effect of ENG amount on radial thermal conductivity of Hydralloy, $LiNH_2$-MgH_2, and $NaAlH_4$. Adapted from.[54b]

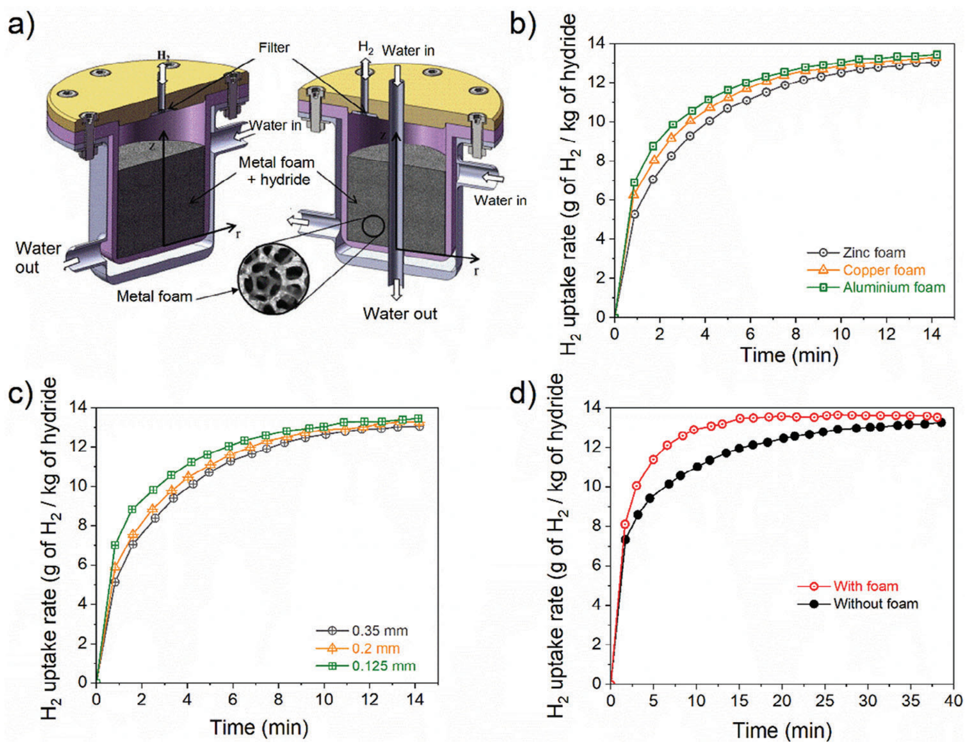


Figure 4. a) Hydrogen storage design with a $LaNi_5$ bed incorporating a metal foam; b) Effect of the metal foam material on the hydrogen uptake rates; c) Effect of the metal foam pore size on the hydrogen absorption rates; and d) Comparison of hydrogen absorption rates of the hydride bed with and without a metal foam. Adapted from.[55a]

Particle & Particle Systems Characterization
www.particle-journal.com

An alternative is to encapsulate the hydrogen storage alloy particles within a metallic "shell".[60] For this, Cu is usually used because copper ions can be easily reduced into metallic Cu at the surface of hydride-forming alloys. In addition, Cu has a high thermal conductivity of 398 W m^{-1} K^{-1}. For example, the formation of a Cu layer at the surface of $LaNi_5$-based particles could be formed by simply immersing the alloy powder in a $CuSO_4$-H_2SO_4 aqueous solution for 90 s.[60a] Upon achieving this, a 500 to 750% increase in thermal conductivity was observed.[60b]

4.2. Improving the Thermal Conductivity of Hydride Storage Tanks

The other approach to improve the performance of hydrogen storage systems is to improve the heat dissipation of the storage system itself. This includes revisiting the tank material, its wall thickness and geometry, and adding fins or introducing heat exchangers.

4.2.1. Tank Materials and Tank Wall Thickness

Ultimately, the effective thermal conductive resistance of hydrogen storage systems should be affected by the thermal conductivity of the vessel material. However, when comparing the effect of using brass, stainless steel, granite, or plexiglass for the tank material, results showed a marginal improvement in hydrogen uptake/release rates when employing brass over the other materials, although brass has a high thermal conductivity of 109 W m^{-1} K^{-1} (Table 4). When compared to the high thermal conductivity of tank material, the thermal conductivity of the $LaNi_5$ bed is much smaller and this dominates the heat transfer process.[61] Other reports led to similar conclusions, e.g. with steel and brass tanks.[32]

The tank wall thickness is also an important factor for both the mechanical strength of the tank and heat management. Investigations of hydrogen storage systems with $LaNi_5$ powder beds and tank walls from 3 to 7 mm indicate that decreasing the tank wall thickness increases the hydrogen charging rate. The tank with the 3.0 mm wall thickness exhibited the shortest filling time of 500 s, while the longest absorption filling time of 3000 s was observed when the tank had a wall thickness of 7.0 mm.[50]

4.2.2. Tank Geometry

Geometry considerations, e.g. tank shape and dimensions, are also important factors to ensure ease of manufacturing, compliance with regulations of pressure vessels, and safety of the storage system. However, these may impact the overall hydrogen uptake/release rates. The shape of the storage tank can also influence factors such as internal pressure distribution. In these considerations, maximizing the internal surface area of the storage tank is essential to increase the contact area between the reactive hydride material and the tank walls to allow for fast heat removal from the hydride bed. Common tank shapes include cylindrical, spherical, and prismatic configurations, each offering advantages and disadvantages in terms of structural integrity, available volume, and thermal management. For example, spherical designs (**Figure 5**b) offer the worst surface area-to-volume ratio theoretically and are hard to design, but they provide the largest volume for hosting the metal hydride bed. Prismatic configurations (Figure 5c), on the other hand, lead to an optimal use of both environment and internal tank space, but are not suitable for high pressures and as such are limited to low-pressure metal hydrides.[62]

A range of simulation studies have been reported for different tank geometries. For example, when comparing a cylindrical vessel, a square-based rectangular vessel, and a rectangle-based rectangular vessel, the latter of higher area-to-volume ratio led to superior wall cooling rates and thus faster hydrogen uptake.[64] A comparison of cylindrical and spherical tanks with a MgH_2 bed and a phase change material for heat management (Figure 5), showed that the time required to reach 80% of hydrogen uptake was 800 and 600 s, respectively. This corresponds to a 22% improvement in storage time upon changing the shape of the tank from cylindrical to spherical.[63] However, in tanks of cylindrical shape, the hydrogen pressure and mechanical stresses are evenly distributed. In addition, heat and mass transfer within the hydride bed is easier to manage especially when it comes to integrating a heat exchanger to the tank.[62] As such, cylindrical tanks remain the most used because these offer a good intermediate solution.

In cylindrical tanks, there is a direct relationship between the length-to-radius ratio (L/r) of the tank and its charge-discharge performance. According to Equation 2, the cross-sectional surface area (A) needs to be maximised, while the distance (d) from the centre of the tank needs to be minimized. This geometry dependence of the hydrogen uptake performance of a cylindrical tank is illustrated in **Figure 6**.

For example, the worst performance is obtained when the L/r ratio is slightly higher than 1. This coincides with the smallest cross-sectional surface area ($L/r = 1$ in Figure 6a), and the maximum distance to the wall ($L/r < 3$ in Figure 6b). Moving off from $L/r \approx 1$ either to the left or the right quickly increases the performance of the tank. Since the cross-sectional surface area increases faster by increasing r instead of L. Disc-shaped cylindrical tanks usually display the best performance (Figure 6d). Similar trends were reported for $LaNi_5$ storage systems with tank aspect ratio L/r of 1.25, 2.4, 5, and 10, where $L/r = 10$ showed the best heat conduction and the fastest hydrogen charge rate. A 17.8% improvement in storage efficiency was achieved with a tank of $L/r = 10$ in comparison with a tank of $L/r = 2.4$.[61] In practical application, a common approach is to combine the performance of $L/r < 1$ and $L/r > 1$ type of tanks. This is usually done by stacking a series of metal hydride pellets inside a long cylindrical tank.[17d]

4.2.3. Adding Fins

One of the preferred options to further improve the heat dissipation in metal-hydride tanks is the use of fins. With the addition of extended surfaces, fins provide high surface contact for heat transfer and shorten thermal exchange paths. Fins also improve the thermal conductivity across the hydride bed and facilitate the extraction of heat from the latter.[65] Fins can take multiple configurations but appear most commonly as a thin metal

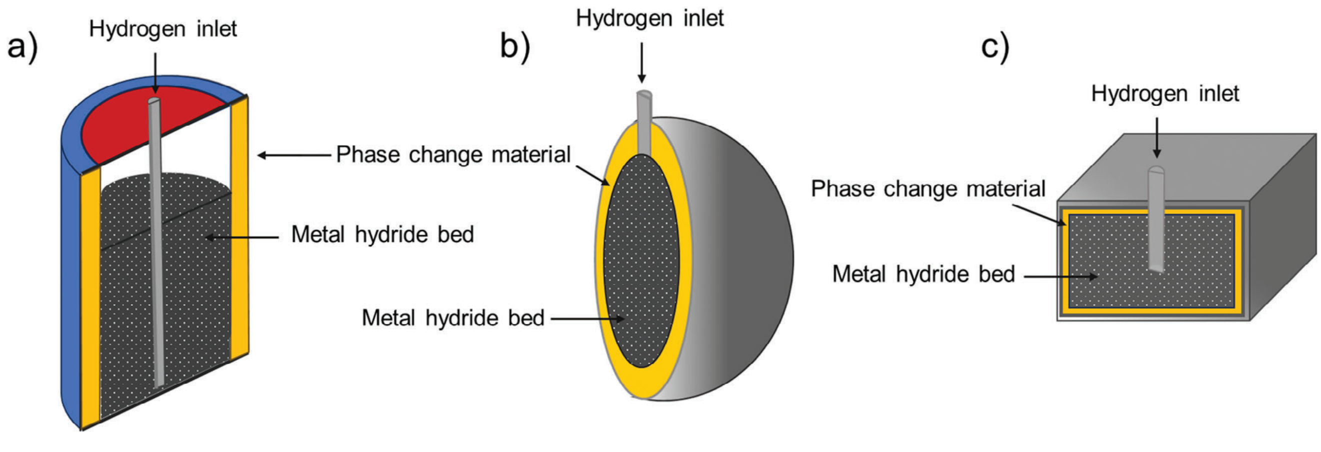


Figure 5. a) Cylindrical tank design; b) Spherical tank design; and c) Prismatic tank design for metal hydride storage tanks with a phase change material for heat management. Adapted from.[63]

plate crossing the bed, either horizontally or vertically. Fins can also take innovative shapes, e.g. tree shape fins or conical shape fins (**Figure 7**).[29,32,66]

Fins not only help dissipate excess heat generated during the exothermic hydrogen absorption process, but they also enhance the heat transfer rate during the hydrogen desorption reaction (**Figure 8a**). This results in a significant acceleration of the absorption and desorption rates of hydrogen within the metal hydride material, leading to faster refuelling or discharge of hydrogen. Design with external fins showed a 10% improvement in the time required for 90% hydrogen uptake in $LaNi_5$ beds.[32] A 60% improvement in hydrogen uptake/release performance was reported for designs with internal fins. Furthermore, tanks designed with internal fins reached lower hydride bed temperatures (80 °C lower than beds without fins).[67] The combination of internal fins with a concentric heat exchanger (Figure 7b) showed an 80% improvement compared to a tank design without fins/heat exchanger.[32]

Fin length, fin thickness, and fin spacing/pitch also affect the heat transfer efficiency of the hydrogen storage tank. Fin diameter is commonly defined as the radial distance of the fin's tip to the circulation fluid tube wall. Studies showed that an increase in fins length leads to an important reduction of storage time, e.g. from 720 s for 1.5 cm fins to 460 s for 2.5 cm fins for $LaNi_5$ storage systems.[32] Improvements of 20–30% in hydrogen uptake rates were also observed with an increase in fin thickness.[68] In addition, tanks employing a lower fin pitch exhibit faster hydrogen absorption rates because of the higher number of fins and the shorter heat transfer path distance.[68a]

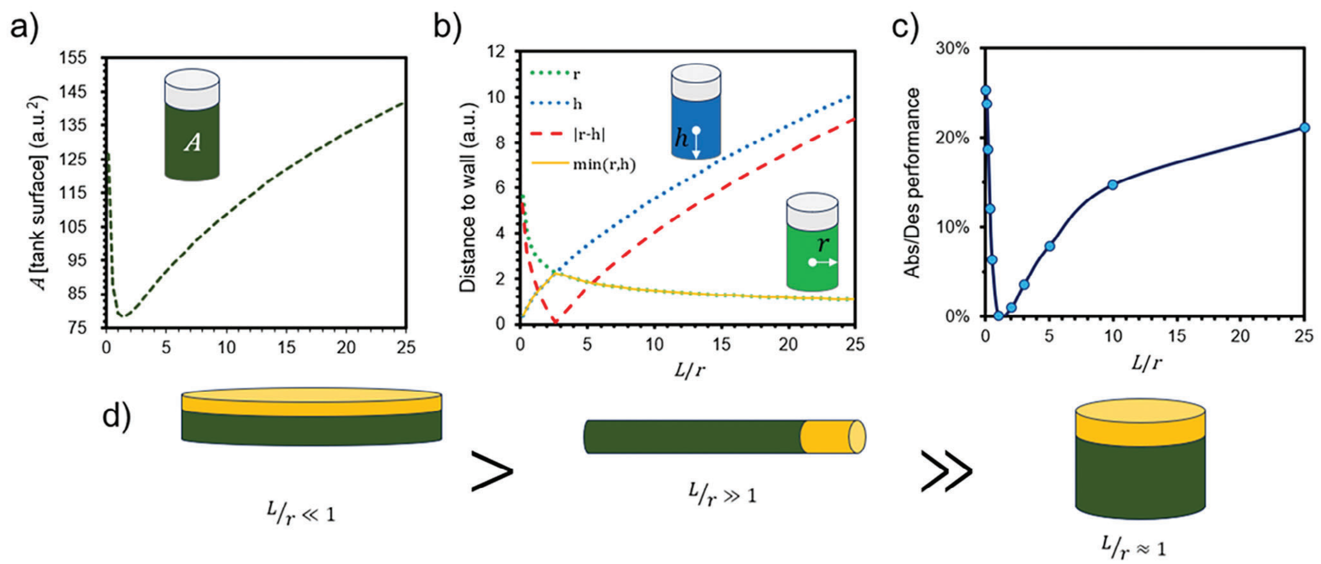


Figure 6. Effect of the length-to-radius ratio (L/r) on the hydrogen uptake performance of cylindrical tanks. a) Contact surface area between the metal hydride bed and tank walls; b) Distance from the center of the metal hydride bed to the wall: vertical (h) and horizontal (r); and c) Hydrogen uptake (Abs)/release (Des) performance of the tank.

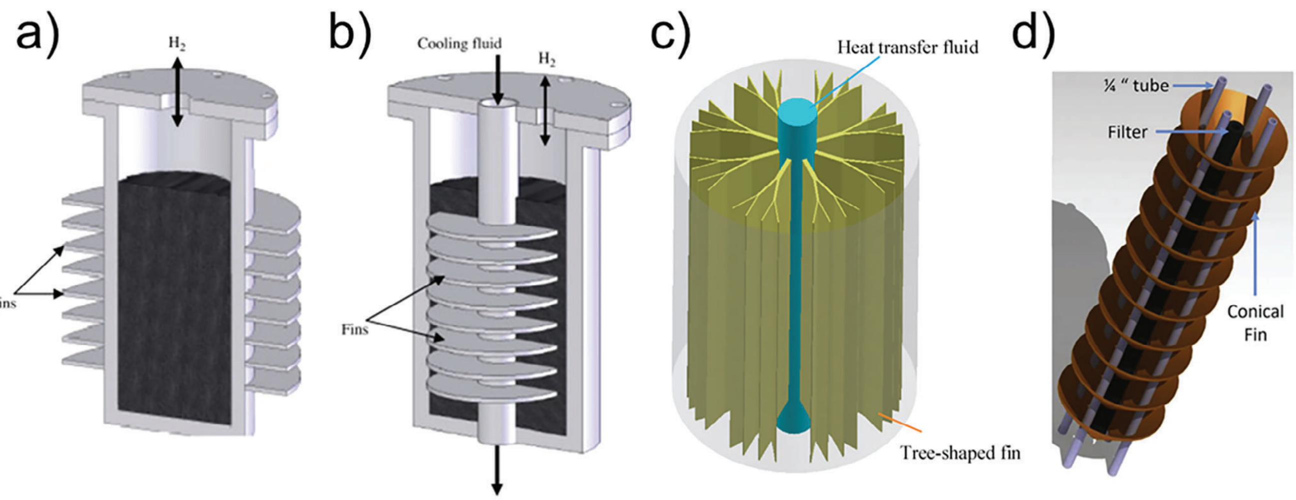


Figure 7. a) Plate fins on the external wall of hydride tank; b) Plate fins inside hydride tank; c) Tree-shape fins; and d) Conical shape fins. Adopted from.[29,32,66b]

Despite the addition of voluminous and potentially heavy metallic parts in the tank, the efficiency of fins at heat dissipation can vastly compensate for the losses of the overall amount of hydrogen stored as a system. Fins occupy a relatively compact space in comparison with heat exchangers. This helps to maintain a good storage capacity of the tank without significantly increasing its footprint. Additionally, fins are generally simpler and less costly to implement, making them a cost-effective solution for temperature control in hydrogen storage tanks. Moreover, the addition of fins can help mitigate the gap in performance between tanks with different L/r ratios (Figure 8b), and this provides an alternative solution when constraints to the dimensions of the tank exist. Fins can be also used in combination with pellets (Figure 8c).

With this configuration, disc-shaped pellets can be stacked along the whole volume of the tank and the use of thermal conducting fins to separate the pellets from each other can help enhance the heat dissipation from the hydride tank.[17d]

4.2.4. Introducing Heat Exchangers

Heat exchangers with suitable heat transfer fluid (HTF) represent another common solution in metal-hydrides hydrogen storage tanks. These help distribute heat uniformly across the metal hydride bed, preventing localized hotspots or cold spots and ensuring consistent performance. Some heat exchangers can be used to recover excess heat generated by fuel cells for example, to compensate for the endothermic nature of the hydrogen release reaction from the metal hydride.[69] Like fins, heat exchangers are presented in multiple configurations, either running through the middle of the bed or built around the tank in a configuration known as a cooling jacket (**Figure 9**a).[55a,68a,70]

An ideal HTF will have a high specific heat to maximize the temperature gradient across the bed (Figure 9c) and be operated at a high flow velocity (Figure 9d). A high flow favors the heat transfer rate by dissipation of the boundary layer in the HTF and the creation of swirls within the flow. High flows also keep the HTF closer to its initial temperature by not allowing enough time for the HTF to reach a higher temperature. Both cases help to maximize the temperature gradient between the wall and the HTF. However, this enhancement of heat transfer rate limits the secondary use of the low-grade heat generated by the hydride tank and as such coupling with other systems.

A cooling jacket is a simple and effective design for heat management of metal hydride tanks. In this case, the cooling jacket is simply built around the tank (**Figure 10**a).[30] Another design is a shell-tube cooling system design whereby several hydride tanks are incorporated within the same water jacket (Figure 10b).[58] Simulation of a 210 kg Ti—Mn-based hydrogen storage system indicated that adding a cooling jacket increased the hydrogen uptake rate by a factor of 2.[30] Compared to this, the shell-tube storage design has also been claimed to lead to a further 50% reduction in hydrogen uptake rates, mainly because of the smaller diameter of the metal hydride tanks.[30]

Another engineering design technique is to place an HTF tube within the metal hydride bed. In this case, the HTF tubes are directly in contact with the hydride bed from which heat can be effectively extracted. Cooling tubes can be straight or coiled (**Figure 11**a). Studies have demonstrated that coiled cooling tubes provide better heat transfer and lead to better hydrogen storage performances. A comparison of embedded straight tubes and embedded coiled tubes in $LaNi_5$ storage systems showed that embedded coiled tubes provided a 73% reduction in charging time compared to a tank without any heat exchanger. This is far superior to the 34% improvement observed with the embedded straight-tube heat exchangers.[31] Similarly, a simulation on a tank containing $MgH_2 + V_2O_5$ reported a reaction completion time reduction of 25% for absorption and 46% for desorption when the embedded cooling tube geometry was changed from straight to helical.[68b] Addition of fins to this was also found to further improve hydrogen uptake/release rates.[68c]

To optimize the hydrogen uptake/release rate upon the use of helical-coil-embedded HTF tubes, the pitch and coil curvature diameter (Figure 11a) are the main two geometric parameters that affect performances. Decreasing the coil pitch, defined as

Particle & Particle Systems Characterization
www.particle-journal.com

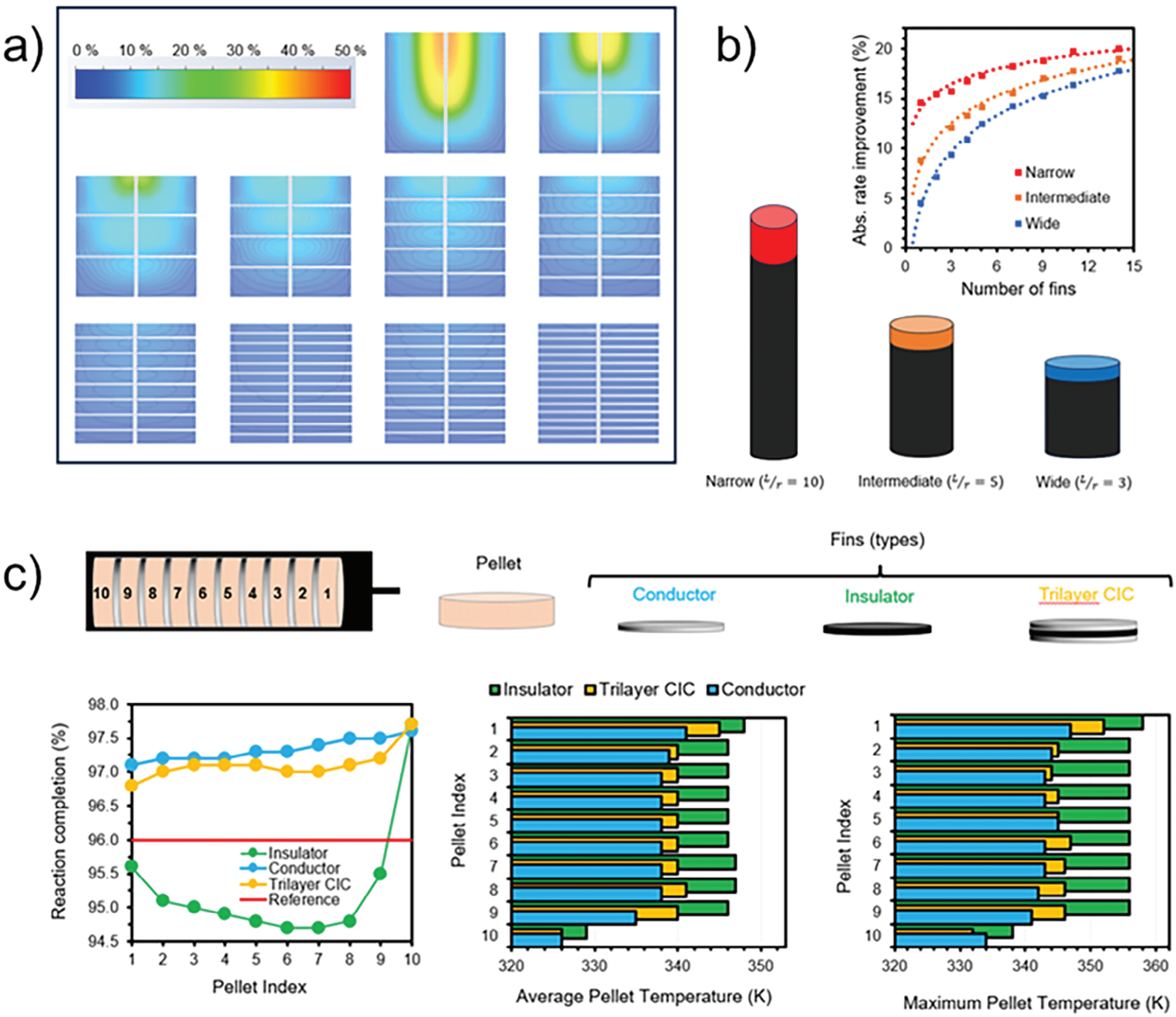


Figure 8. Effect of the introduction of fins in a hydride tank. a) Hydrogen release rate of 10 tanks of equal volume but with a different number of fins (varying from 0 to 9); b) Hydrogen uptake rate when the number of fins is increased from 0 to 15 in three tanks of equal volume but different L/r ratio; c) Performance and hydride bed temperature of a tank with 10 pellets and 9 fins. Three different types of fins were used: conductor/metal (blue), insulator (green), and conductor–insulator–conductor (CIC – orange). Reference tank (red): same volume, no fins, and $L/r = 3$.

the axial distance between two consecutive coils, will result in a higher heat transfer contact surface area because of the longer total length of the embedded HTF tube. For example, the hydrogen uptake rate was reported to be higher with an 8 mm helical pitch compared to the tanks employing pitches of 10, 12, and 14 mm (Figure 11b,c)[71] Increasing the helical curvature diameter will also increase the area for heat exchange between HTF and the hydride bed (Figure 11d,e).[71,72] The HTF tube diameter also has a positive influence on the storage performance of metal hydrides. Increasing the HTF tube diameter from 3.0 to 6.0 mm has been reported to increase the hydrogen uptake rate by 50%.[71,72] It is important to note that by optimizing these geometric parameters (lower pitch, higher helical curvature diameter, and higher HTF tube diameter), both the mass and volume occupied by the heat exchanger increase, and this impacts the overall hydrogen storage capacity of the system. Therefore, without a system that can re-use the heat generated by the metal hydride bed upon hydrogen uptake/release, the use of heat exchangers is not justified. These add complexity to the design and operation of hydrogen storage systems, and additional maintenance issues. Heat exchangers occupy space within the storage tank, potentially reducing the volume available for hydrogen storage. Despite their efficiency, heat exchangers may also result in some heat losses during the transfer process, leading to a reduced overall system efficiency. The implementation of heat exchangers also increases the cost of the storage system because of the need for specialized materials and components. Before considering the use of any heat exchanger in a hydride tank design, the metal hydride

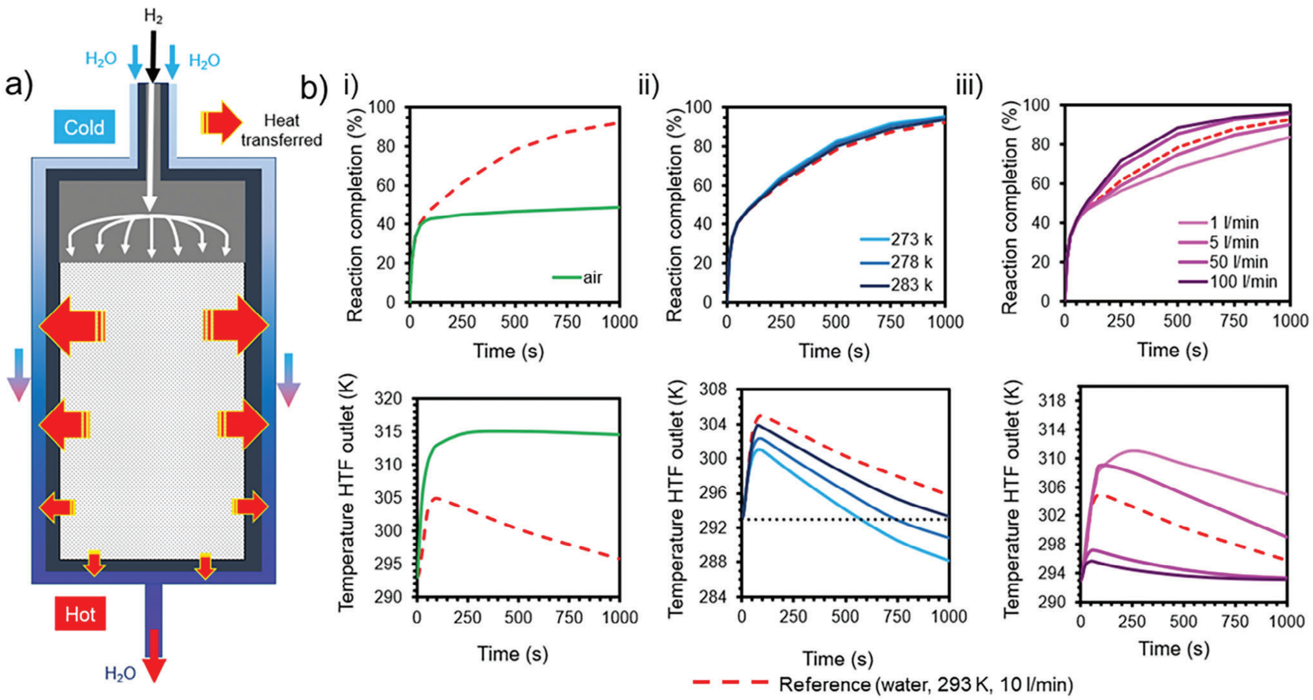


Figure 9. Effect of heat exchangers on the performance of metal hydride tanks. a) Heat exchanger configuration around metal hydride tanks as a jacket; and b) Hydrogen uptake reaction completion rate and HTF outlet temperature: i) versus time with air or water as HTF, ii) at different HTF inlet temperatures, iii) versus time with HTF at different flowrate.

should first be optimized for its operability against a specific application. Additional heat exchangers may then be combined into the system for marginal corrections of heat compensations.

5. Conclusions

Metal hydrides in powder form have a low thermal conductivity, which leads to poor charging/discharging rates for reported hydrogen storage systems. As such modeling tools based on mass balance, momentum conservation, and energy efficiency with the capability to set initial and boundary conditions are powerful tools to not only simulate hydrogen storage systems but also test new tank design concepts.

The thermal conductivity of hydride beds is usually improved by compacting the hydride-forming alloy with additives of high thermal conductivity including expanded natural graphite or metallic foams. Both methods have been reported to be effective in improving the hydride bed thermal conductivity. However, compacted pellets are often the preferred solution because compaction helps to increase the density of the metal hydride in the tank while improving its thermal conductivity. However, the amount of additive and the level of compaction must be well controlled to allow sufficient porosity for fast hydrogen diffusion, while enabling the breathing of the hydride pellet, i.e., its expansion and contraction, upon the uptake and release of hydrogen.

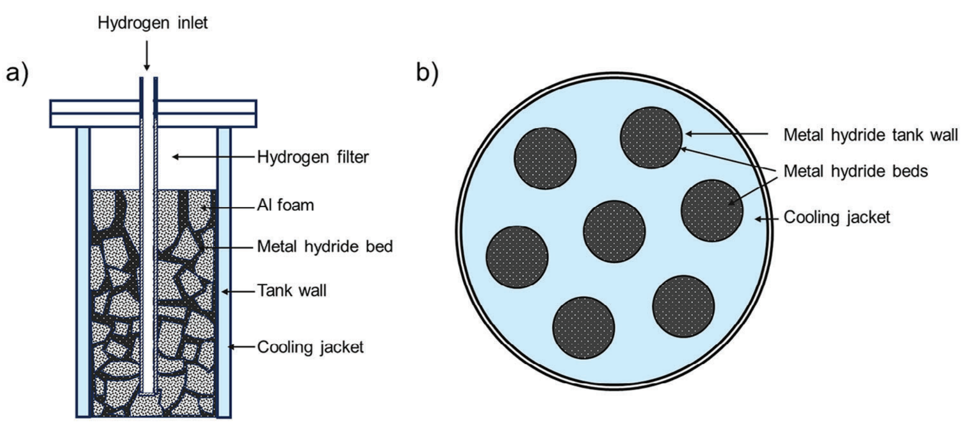


Figure 10. a) Common cooling jacket for metal hydride bed; and b) Cooling jacket in a shell-tube type storage tank. Adapted from.[29,61]

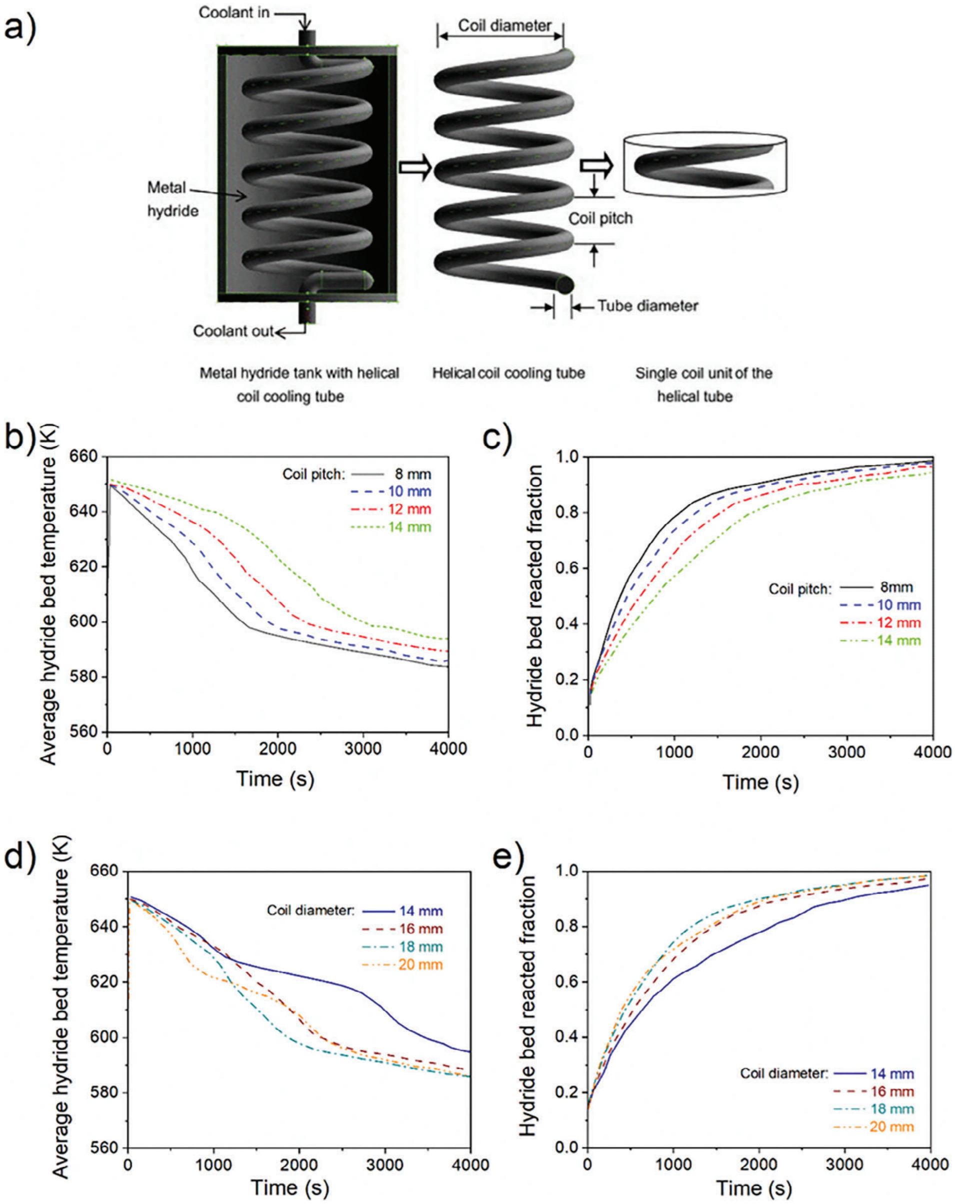


Figure 11. a) Coiled cooling tube; b) Effect of coil pitch on hydride bed temperature; c) Effect of coil pitch on bed reaction rate; d) Effect of coil diameter on hydride bed temperature; and d) Effect of coil diameter on bed reaction fractions. Adopted from.[69,71]

Once the hydride bed thermal conductivity and hydrogen sorption rates have been balanced, the next level of optimization resides in the design of the storage tank to allow for rapid dissipation of the heat generated by the exo(endo)thermic nature of the hydrogen (abs)desorption reaction in metal hydrides. In this respect, cylindrical tanks are still considered the best choice compared to prismatic or spherical storage vessels, because a cylindrical design allows for operation at higher hydrogen pressures. A range of pipe sizes from which cylindrical hydrogen storage tanks can be fabricated are also commercially available, and this facilitates the design and construction of metal hydride tanks.

Tank materials and wall thickness can be thought of as the main factors affecting the rate of heat dissipation from the metal hydride bed within the tank. However, because the rate-determining factor of the hydrogen uptake/release rate often resides in the low thermal conductivity of hydride beds, the tank wall thickness and material type are not so important. As such disc-like tanks and tubular tanks are proposed to have better performance because this leads to a larger exposed surface area through which heat can be dissipated. As such a compromise is often to stack pellets of the hydride within a long tubular tank. Additionally, heat removal from hydride tanks can be enhanced by adding fins or introducing heat exchangers. Fins can be added internally or externally to the tank. Internal fins are often preferred because of the better contact with the hydride bed and thus the greater heat dissipation. In this case, further improvement in thermal transfer can be adjusted by increasing the fin length and thickness and by lowering the fin pitch. Heat exchangers are more cumbersome systems, and these usually take the form of an external cooling jacket or internal cooling tube (straight or coiled). Once again, internal cooling tubes show better heat transfer performance than an external cooling jacket because of the direct contact with the hydride bed. However, this leads to an overall hydride tank system of lower hydrogen storage capacity and an increase in cost owing to the complicated designs of internal heat exchange tubes resistant to hydrogen embrittlement and capable of withstanding high hydrogen pressures. As such a popular design is to arrange bundles of tubular metal hydride tanks within a larger cooling vessel where all the tubular hydride tanks are cooled at the same time. For the current state of the art, it is thus apparent that heat management of metal hydride beds remains a fine art in balancing contradictory requirements- One wishes to store and transport hydrogen and not stainless steel. In this respect, novel ideas on ways to develop a passive heat management system are required, as well as the development of alternative tank and heat exchanger materials capable of withstanding hydrogen pressure and across reasonable temperature windows of 20 to 150 °C. Resistance of these new materials to hydrogen leakage and embrittlement is a must. In this respect, simulation tools should evolve to predictive modes to guide not only the material choice of tanks but also their designs.

Acknowledgements

The authors acknowledge the support of funding from the ARC Training Centre for the Global Hydrogen Economy (GlobH2E) (IC200100023) as well as the use of the National Computational Infrastructure (NCI) which was supported by the Australian Government and accessed through the Sydney Informatics Hub HPC Allocation Scheme, which is supported by the Deputy Vice-Chancellor (Research), University of Sydney.

Conflict of Interest

The authors declare no conflict of interest.

Keywords

heat transfers, hydrogen storage, metal hydrides, tank designs, thermal management

Received: July 16, 2024
Revised: September 2, 2024
Published online: October 22, 2024

[1] P. Modi, K. F. Aguey-Zinsou, *Front. Energy Res.* **2021**, *9*, 616115.
[2] P. Lv, in *Toward Hydrogen Infrastructure* (Eds: D. Jaiswal-Nagar, V. Dixit, S. Devasahayam), Elsevier, Amsterdam **2024**, 165.
[3] I. Pielecha, D. Engelmann, J. Czerwinski, J. Merkisz, *Rail Vehicles/Pojazdy Szynowe* **2022**, *10*, 147725.
[4] Y. Bi, Y. Ju, *Energy* **2022**, *252*, 124047.
[5] N. A. A. Rusman, M. Dahari, *Int. J. Hydrogen Energy* **2016**, *41*, 12108.
[6] L. Guo, Z. Wu, H. Wang, H. Yan, F. Yang, G. Cheng, Z. Zhang, *Chem. Eng. J.* **2023**, *455*, 140689.
[7] J. Barale, F. Nastro, D. Violi, P. Rizzi, C. Luetto, M. Baricco, *Int. J. Hydrogen Energy* **2023**, *48*, 34105.
[8] B. Sakintuna, F. Lamari-Darkrim, M. Hirscher, *Int. J. Hydrogen Energy* **2007**, *32*, 1121.
[9] G. G. Lucas, W. L. Richards, *Int. J. Hydrogen Energy* **1984**, *9*, 225.
[10] a) L. Tong, C. Yuan, T. Yang, Y. Yuan, R. Chahine, J. Xiao, *Therm. Eng.* **2023**, *43*, 102812; b) J. Zhu, X. Lin, L. Lv, M. Li, Q. Luo, V. N. Kudiiarov, W. Liu, H. Leng, X. Han, Z. Ma, *J. Mater. Sci. Technol.* **2024**, *203*, 66.
[11] S. S. Mohammadshahi, E. M. Gray, C. J. Webb, *Int. J. Hydrogen Energy* **2016**, *41*, 3470.
[12] a) U. Mayer, M. Groll, W. Supper, *J. Less-Common Met.* **1987**, *131*, 235; b) I. A. El Osery, *Int. J. Hydrogen Energy* **1983**, *8*, 191; c) S. Da-Wen, D. Song-Jiu, *J. Less-Common Met.* **1989**, *155*, 271.
[13] S. S. Mohammadshahi, T. Gould, E. M. Gray, C. J. Webb, *Int. J. Hydrogen Energy* **2016**, *41*, 3537.
[14] V. Kudiiarov, R. Elman, N. Pushilina, N. Kurdyumov, *Materials* **2023**, *16*, 4891.
[15] a) K. Faraj, M. Khaled, J. Faraj, A. Faraj, C. Castelain, *J. Phys. Conf. Ser.* **2024**, *2754*, 012030; b) G. Miao, P. Li, C. Liu, Y. Liu, H. Zhang, F. Lin, X. Qu, *Sustainable Energy Fuels* **2023**, *7*, 2025.
[16] G. B. Arfken, D. F. Griffing, D. C. Kelly, J. Priest, in *International Edition University Physics* (Eds: G. B. Arfken, D. F. Griffing, D. C. Kelly, J. Priest), Academic Press, USA **1984**, 430.
[17] a) S. B. Nasrallah, A. Jemni, *Int. J. Hydrogen Energy* **1997**, *22*, 67; b) A. Jemni, S. B. Nasrallah, *Int. J. Hydrogen Energy* **1995**, *20*, 43; c) A. Freni, F. Cipitì, G. Cacciola, *Int. J. Hydrogen Energy* **2009**, *34*, 8574; d) Y. Ye, J. Lu, J. Ding, W. Wang, J. Yan, *Appl. Energy* **2020**, *278*, 115682.
[18] Z. Zhou, J. Zhang, J. Ge, F. Feng, Z. Dai, *Int. J. Hydrogen Energy* **1994**, *19*, 269.
[19] A. Y. Esayed, D. O. Northwood, *Int. J. Hydrogen Energy* **1997**, *22*, 77.
[20] T. Nishizaki, K. Miyamoto, K. Yoshida, *J. Less-Common Met.* **1983**, *89*, 559.
[21] A. Atangana, in *Fractional Operators with Constant and Variable Order with Application to Geo-Hydrology* (Ed: A. Atangana), Academic Press, USA **2018**, 15.
[22] J. F. Richardson, J. H. Harker, J. R. Backhurst, in *Chemical Engineering*, 5th Ed. (Eds: J. F. Richardson, J. H. Harker, J. R. Backhurst), Butterworth-Heinemann, Oxford **2002**, 191.
[23] M. P. Boyce, in *Gas Turbine Engineering Handbook*, 4th Ed., (Ed: M. P. Boyce), Butterworth-Heinemann, Oxford **2012**, 139.
[24] A. Morega, M. Morega, A. Dobre, in (Eds: A. Morega, M. Morega, A. Dobre), Academic Press, USA **2021**, 1.
[25] P. Chippar, S. D. Lewis, S. Rai, A. Sircar, *Int. J. Hydrogen Energy* **2018**, *43*, 8007.
[26] A. Karmakar, A. Mallik, N. Gupta, P. Sharma, *Int. J. Hydrogen Energy* **2021**, *46*, 5495.
[27] X. Lin, W. Xie, Q. Zhu, H. Yang, Q. Li, *Chem. Eng. J.* **2021**, *421*, 127844.
[28] M. Afzal, P. Sharma, *Int. J. Hydrogen Energy* **2021**, *46*, 13116.
[29] S. Chandra, P. Sharma, P. Muthukumar, S. S. V. Tatiparti, *Int. J. Hydrogen Energy* **2020**, *45*, 8794.
[30] M. Afzal, P. Sharma, *Int. J. Hydrogen Energy* **2018**, *43*, 13356.
[31] L. Tong, J. Xiao, T. Yang, P. Bénard, R. Chahine, *Int. J. Hydrogen Energy* **2019**, *44*, 15907.
[32] F. Askri, M. Ben Salah, A. Jemni, S. Ben Nasrallah, *Int. J. Hydrogen Energy* **2009**, *34*, 897.
[33] S. Mellouli, F. Askri, H. Dhaou, A. Jemni, S. B. Nasrallah, *Int. J. Hydrogen Energy* **2009**, *34*, 3005.
[34] M. Raju, S. Kumar, *Int. J. Hydrogen Energy* **2011**, *36*, 1578.
[35] M. Bhouri, J. Goyette, B. J. Hardy, D. L. Anton, *Int. J. Hydrogen Energy* **2011**, *36*, 6723.
[36] S. Jana, P. Muthukumar, *J. Energy Storage* **2021**, *39*, 102612.
[37] C. N. Ranong, M. Höhne, J. Franzen, J. Hapke, G. Fieg, M. Dornheim, N. Eigen, J. M. Bellosta von Colbe, O. Metz, *Chem. Eng. Technol.* **2009**, *32*, 1154.

[38] HyCARE Project, https://hycare-project.eu/ (accessed: September 2024).
[39] G.K.N. Hydrogen, https://www.gknhydrogen.com/ (accessed: September 2024).
[40] D. A. Corrigan, Ovonic Metal Hydride Technologies for Photovoltaic Energy Storage Applications, https://www1.eere.energy.gov/solar/pdfs/sda_12corrigan.pdf (accessed: September 2024).
[41] A. Franz, H. Zoz, in *The 4th International Conference on Powder Metallurgy in Asia*, Hsinchu, Taiwan **2017**.
[42] L. K. Heung, On-Board Hydrogen Storage System using Metal Hydride, https://www.osti.gov/servlets/purl/503505 (accessed: September 2024).
[43] A. Nakano, T. Maeda, H. Ito, T. Motyka, J. M. Perez-Berrios, S. Greenway, *Energy Procedia* **2012**, *29*, 463..
[44] a) K. J. Kim, B. Montoya, A. Razani, K. H. Lee, *Int. J. Hydrogen Energy* **2001**, *26*, 609; b) Z. Abdin, C. J. Webb, E. M. Gray, *Int. J. Hydrogen Energy* **2018**, *43*, 5048.
[45] Y. Yang, X. Mou, Z. Zhu, Z. Bao, *Int. J. Hydrogen Energy* **2021**, *46*, 19467.
[46] a) H. Q. Nguyen, B. Shabani, *Int. J. Hydrogen Energy* **2021**, *46*, 31699; b) E. Hahne, J. Kallweit, *Int. J. Hydrogen Energy* **1998**, *23*, 107; c) Z. Dehouche, N. Grimard, F. Laurencelle, J. Goyette, T. K. Bose, *J. Alloys Compd.* **2005**, *399*, 224.
[47] H. Wang, A. K. Prasad, S. G. Advani, *Int. J. Hydrogen Energy* **2012**, *37*, 290.
[48] J. Jepsen, C. Milanese, J. Puszkiel, A. Girella, B. Schiavo, G. A. Lozano, G. Capurso, J. M. Bellosta von Colbe, A. Marini, S. Kabelac, M. Dornheim, T. Klassen, *Energies* **2018**, *11*, 1081.
[49] U. Elmas, F. Bedir, M. Kayfeci, *Int. J. Hydrogen Energy* **2018**, *43*, 10741.
[50] F. Elhamshri, M. Kayfeci, U. Matik, S. Alous, *Int. J. Hydrogen Energy* **2020**, *45*, 34956.
[51] X. Lin, D. Sun, S. Chen, Q. Zhu, H. Leng, Q. Li, *Appl. Therm. Eng.* **2019**, *161*, 114129.
[52] a) J. Datsko, L. Hartwig, B. McClory, *J. Mater. Eng. Perform.* **2001**, *10*, 718; b) R. M. Christensen, Physical Ductility of the Elements, https://failurecriteria.com/physicalductilit.html (accessed: September 2024).
[53] a) M. Ron, E. Bershadsky, Y. Josephy, *J. Less-Common Met.* **1991**, *172*, 1138; b) M. Ron, E. Bershadsky, Y. Josephy, *Int. J. Hydrogen Energy* **1992**, *17*, 623; c) E. Bershadsky, Y. Josephy, M. Ron, *J. Less-Common Met.* **1989**, *153*, 65.
[54] a) C. Pohlmann, L. Röntzsch, T. Weißgärber, B. Kieback, *Int. J. Hydrogen Energy* **2013**, *38*, 1685; b) C. Pohlmann, L. Röntzsch, F. Heubner, T. Weißgärber, B. Kieback, *J. Power Sources* **2013**, *231*, 97; c) C. Pohlmann, L. Röntzsch, S. Kalinichenka, T. Hutsch, B. Kieback, *Int. J. Hydrogen Energy* **2010**, *35*, 12829; d) C. Pohlmann, L. Röntzsch, J. Hu, T. Weißgärber, B. Kieback, M. Fichtner, *J. Power Sources* **2012**, *205*, 173.
[55] a) S. Mellouli, H. Dhaou, F. Askri, A. Jemni, S. B. Nasrallah, *Int. J. Hydrogen Energy* **2009**, *34*, 9393; b) S. Ferekh, S. Kyoung, H. Ju, *Fusion Eng. Des.* **2016**, *109*, 884; c) X. S. Bai, W. W. Yang, Y. J. Yang, K. R. Zhang, F. S. Yang, *Int. J. Hydrogen Energy* **2022**, *47*, 35340.
[56] F. Laurencelle, J. Goyette, *Int. J. Hydrogen Energy* **2007**, *32*, 2957.
[57] M. Lototskyy, B. Satya Sekhar, P. Muthukumar, V. Linkov, B. G. Pollet, *J. Alloys Compd.* **2015**, *645*, S117.
[58] S. Shaji, G. Mohan, *J. Alloys Compd.* **2018**, *735*, 2675.
[59] M. L. Tsai, T. S. Yang, *Int. J. Hydrogen Energy* **2010**, *35*, 11052.
[60] a) G. Atalmis, S. Toros, B. Timurkutluk, Y. Kaplan, *Int. J. Hydrogen Energy* **2024**, *53*, 647; b) G. Atalmis, M. Demiralp, N. Yelegen, Y. Kaplan, *Int. J. Hydrogen Energy* **2023**, *48*, 23067; c) J. K. Kim, I. S. Park, K. J. Kim, K. Gawlik, *Int. J. Hydrogen Energy* **2008**, *33*, 870.
[61] C. S. Wang, J. Brinkerhoff, *Int. J. Hydrogen Energy* **2021**, *46*, 24256.
[62] A. Mazzucco, M. Dornheim, M. Sloth, T. R. Jensen, J. O. Jensen, M. Rokni, *Int. J. Hydrogen Energy* **2014**, *39*, 17054.
[63] S. Mellouli, N. Ben Khedher, F. Askri, A. Jemni, S. B. Nasrallah, *Appl. Therm. Eng.* **2015**, *90*, 674.
[64] J. Nam, J. Ko, H. Ju, *Appl. Energy* **2012**, *89*, 164.
[65] T. L. Bergman, *Fundamentals of heat and mass transfer*, John Wiley & Sons, Hoboken, NJ **2011**.
[66] a) S. N. Nyamsi, F. Yang, Z. Zhang, *Int. J. Hydrogen Energy* **2012**, *37*, 16078; b) X. S. Bai, W. W. Yang, X. Y. Tang, F. S. Yang, Y. H. Jiao, Y. Yang, *Energy* **2021**, *220*, 119738; c) Y. Zhuo, S. Jung, Y. Shen, *Energy Fuels* **2021**, *35*, 10908.
[67] S. Ferekh, G. Gwak, S. Kyoung, H. Kang, M. Chang, S. Yun, Y. Oh, W. Kim, D. Kim, T. Hong, H. Ju, *Int. J. Hydrogen Energy* **2015**, *40*, 14540.
[68] a) S. Mellouli, F. Askri, H. Dhaou, A. Jemni, S. B. Nasrallah, *Int. J. Hydrogen Energy* **2010**, *35*, 1693; b) I. Ayub, M. S. Nasir, Y. Liu, A. Munir, Z. Wu, F. Yang, Z. Zhang, *Int. J. Hydrogen Energy* **2020**, *45*, 31612; c) A. Singh, M. P. Maiya, S. S. Murthy, *Int. J. Hydrogen Energy* **2015**, *40*, 9733.
[69] S. A. Cetinkaya, T. Disli, G. Soyturk, O. Kizilkan, C. O. Colpan, *Energies* **2023**, *16*, 341.
[70] a) R. U. Urunkar, S. D. Patil, *Int. J. Hydrogen Energy* **2021**, *46*, 19486; b) A. H. Eisapour, M. Eisapour, P. Talebizadehsardari, G. S. Walker, *J. Energy Storage* **2021**, *36*, 102443.
[71] A. H. Eisapour, A. Naghizadeh, M. Eisapour, P. Talebizadehsardari, *Int. J. Hydrogen Energy* **2021**, *46*, 14478.
[72] S. Tiwari, P. Sharma, *J. Energy Storage* **2021**, *41*, 102845.

Wei Liu completed his Ph.D. degree in Chemical Engineering at the University of New South Wales in 2015. Currently he works as a postdoctoral researcher in School of Chemistry at the University of Sydney. Dr. Liu's research interests include hydrogen storage materials, hydrogen production, nanomaterials and catalysis. He is now working on the exploration of alloys for high-capacity hydrogen storage.

Joseph Almar Tupe is currently pursuing his Ph.D. in Chemistry under the supervision of Prof. Francois Aguey-Zinsou. Joeaims to develop realistic and reliable techno-economic hydrogen system models.Joe has eight years of experience in the renewable energy industry, including project development for geothermal, solar, and connection projects. He has abachelor's degree in chemical engineering and a master's degree in energy development.

Kondo-Francois Aguey-Zinsou is heading the MERLin research lab at the University of Sydney. Francois is an expert in hydrogen technologies. His group has been working for many years on the properties of light metals and their hydrides for hydrogen applications. He is also developing technologies for enabling novel fuel cell and electrolyzer architectures and advanced catalysts for hydrogen combustion. Francois Aguey-Zinsou isco-founder and CTO of several hydrogen start-ups.

